Whitening & Anti-Oxidation

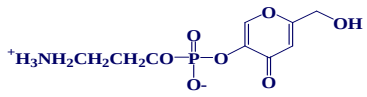
Kojic Acid derivative

*Trade Name : Whitegen

INCI Name : Aminopropyl Kojyl Phosphate

Introduction

- Stabilized Kojic acid derivative for whitening
- It is Aminopropyl Kojyl phosphate, obtained from the condensation reaction of Kojic acid and 3-aminopropyl phosphoric acid.
- Safe and efficient skin whitening agent
- Strong removing action of hyperpigmentation



Kojyl 3-aminopropyl phosphate

<Whitegen molecular formula>

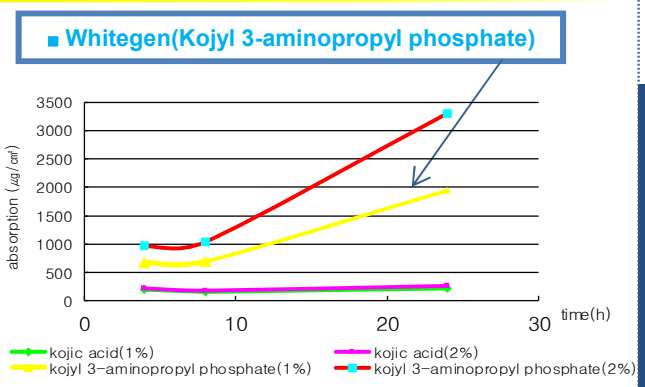
Specification

Appearance	White powder
pH(1→100)	4.0 ~ 6.0
Solubility	Water Soluble
Dosage	0.5~2%
Applicable to skincare and body care products	

Advantages of Whitegen

- **High stability**
  Retain >90% stability for 30 days in water
- **Skin Permeability**
  Permeable >10 times higher when compared to **Kojic acid alone**
- **Action mechanism**
  Kojic acid and 3-aminopropyl acid reaction helps to penetrations into skin

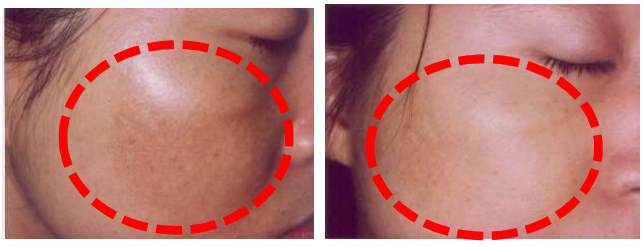
Whitening Efficacy (1) Skin Penetration



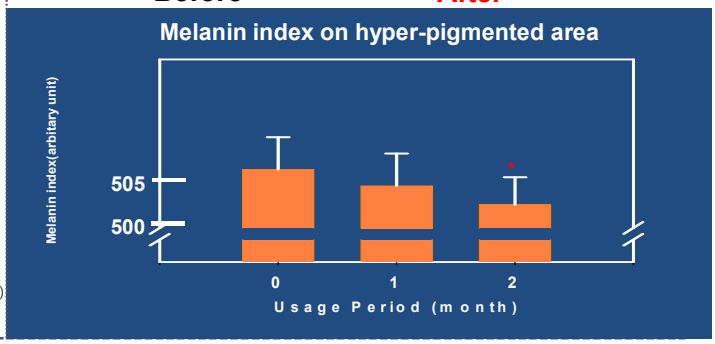
Whitening Efficacy (2) Clinical studies

- Subjects : 30 healthy females having serious hyper-pigmentation (melasma, liver sot, age spot and freckles) on face, mean age = 42.5 year, Evaluated with Mexameter
- Test periods: 2 months
- Sample : 2% kojyl-APPA lotion

(Done by Dept. of Dermatology, Seoul Nat'l Univ.)



Before **After**



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